HESTIA

A UNIFIED CAMPUS RECRUITMENT, CAREER ACCELERATION & SECURE CREDENTIAL VERIFICATION SYSTEM

MICRO-PROJECT REPORT

submitted by

Nripan Satheesh (MBC25CA063)

Muhammed Afrithkhan (MBC25CA057)

to

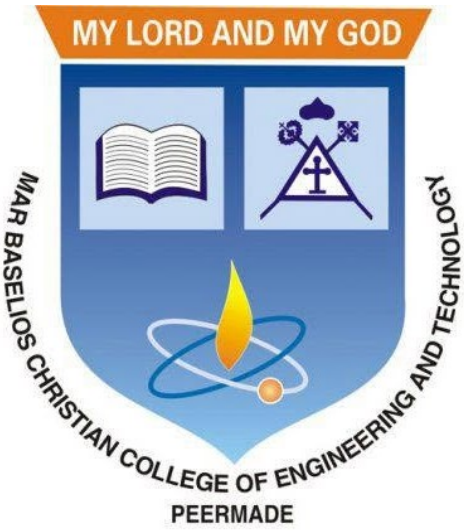
APJ Abdul Kalam Kerala Technological University

in partial fulfilment of the requirements for the award of the Degree of

Bachelor of Technology

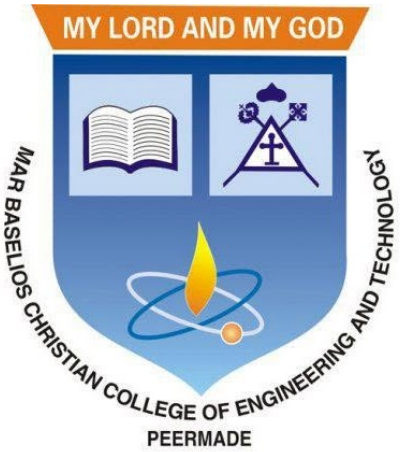
in

Computer Science and Engineering (AI)



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)
MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY, PEERMADE - 685 531

OCTOBER, 2026



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)
MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY, PEERMADE

CERTIFICATE

This is to certify that the micro project report entitled **"HESTIA"** submitted by **Nripan Satheesh (MBC25CA063)** and **Muhammed Afrithkhan (MBC25CA057)** to APJ Abdul Kalam Kerala Technological University in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Technology in Computer Science and Engineering (AI)** is a bonafide record of the micro project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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3.4 SYSTEM ARCHITECTURE DIAGRAM

The high-level enterprise architecture diagram illustrating the interaction between the Presentation Layer, API Gateway, Verification Facade, Frontier AI Core, and Relational Persistence Tier is presented below:

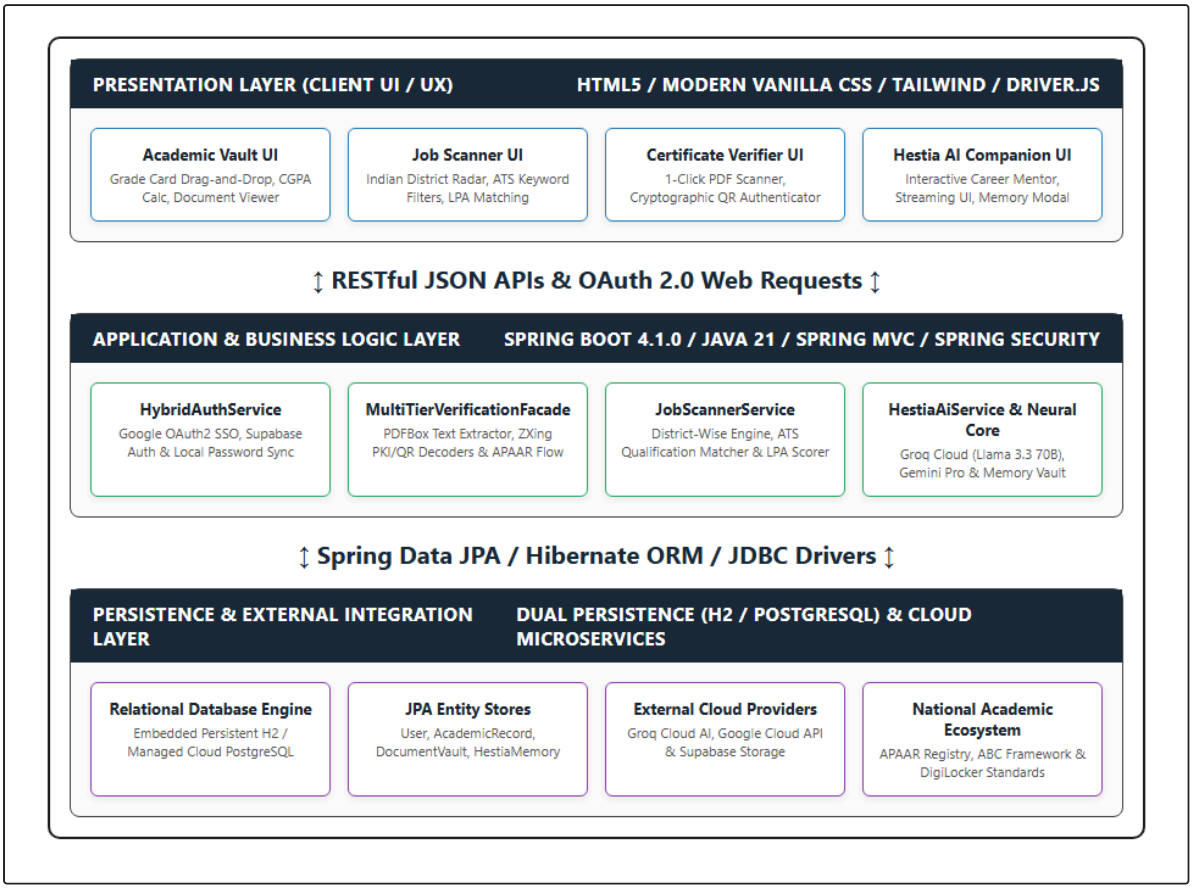


Fig 3.1: Enterprise Architecture of Hestia Platform

As depicted in Fig 3.1, client requests transit the secure API Gateway before dispatching to dedicated service facades, ensuring robust data isolation and zero single points of failure.

3.5 CONCEPTUAL RELATIONAL DATABASE DESIGN

The persistence model of Hestia is designed in Third Normal Form (3NF) to eliminate data redundancy and preserve referential integrity across student records, academic grade cards, verification logs, and AI conversation histories.

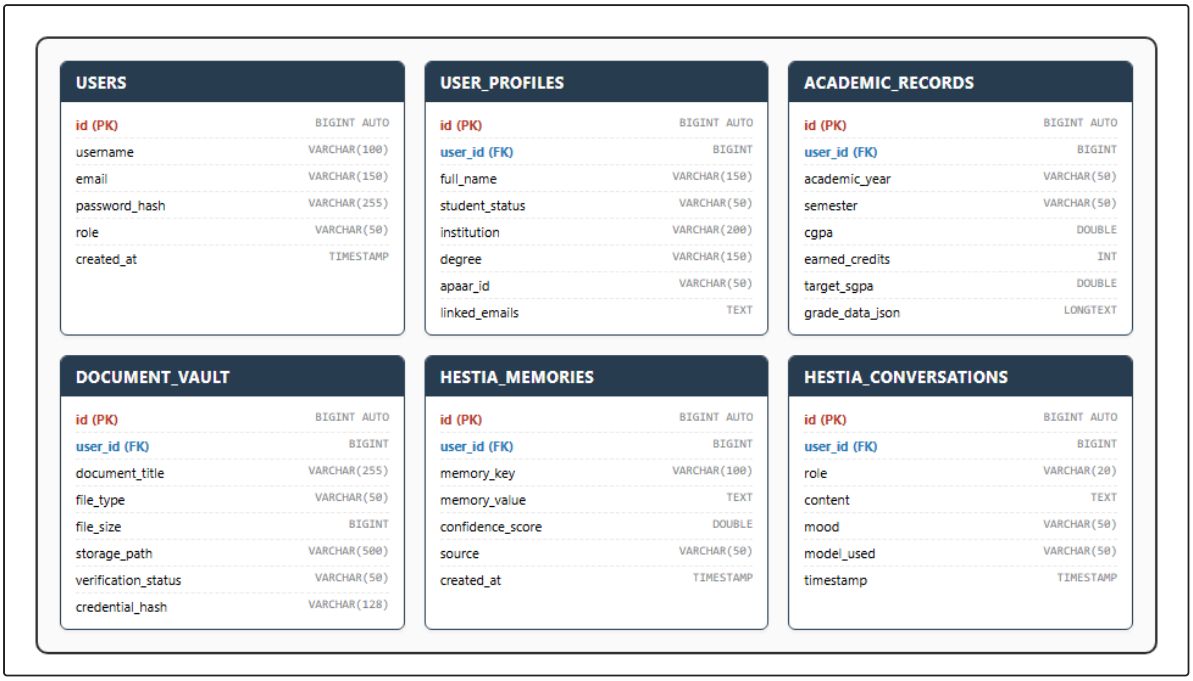


Fig 3.2: Relational Database Schema & Entity Relationships

Primary relational entities include Users, AcademicRecords, VerificationAudits, JobPostings, and AIMemoryVault, maintaining strict foreign key cascade rules and optimized B-Tree indexation.

CHAPTER-6
SCREENSHOTS

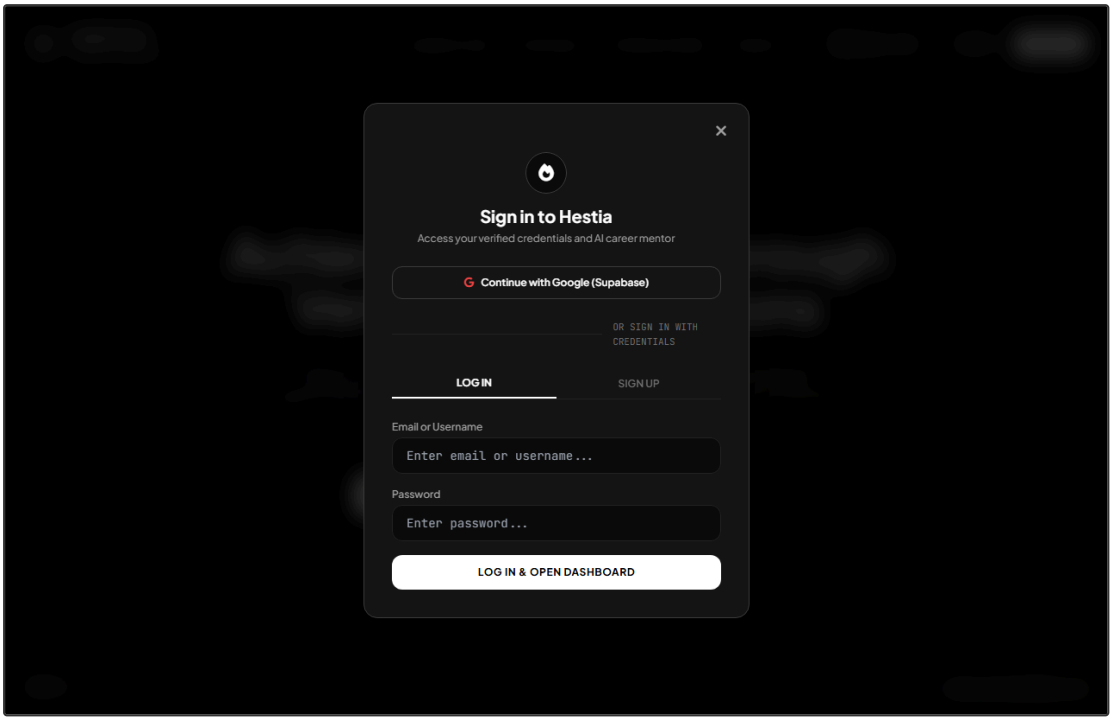


Fig 6.1: Hybrid Authentication Modal (Google SSO & Password Sync)

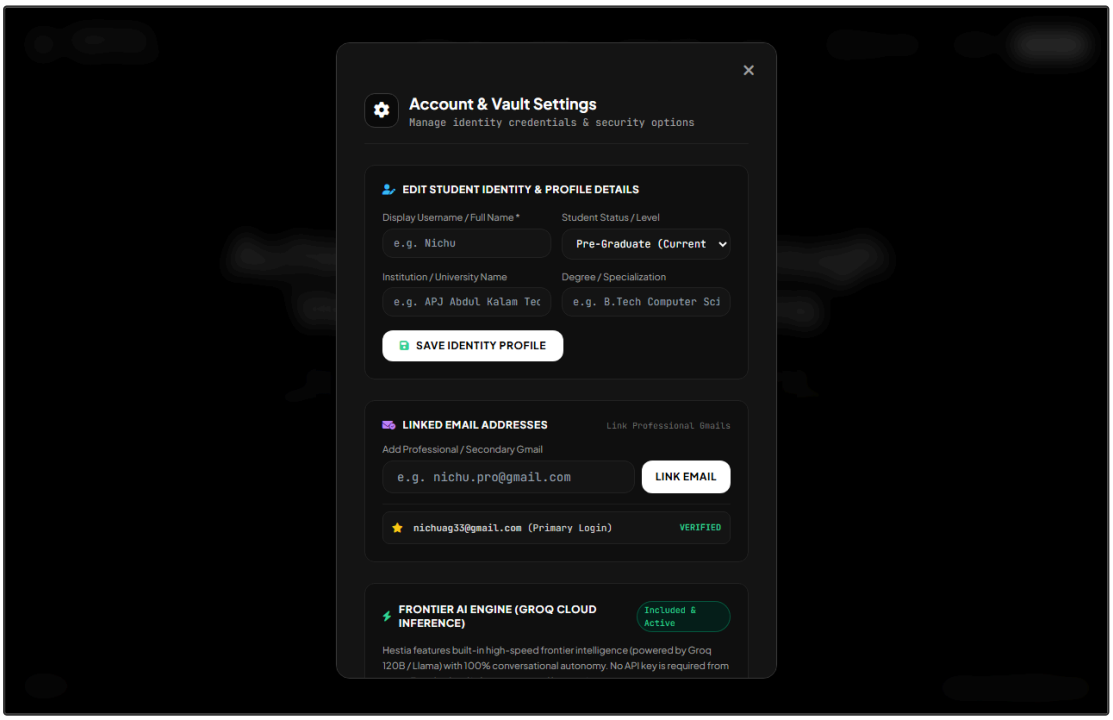


Fig 6.2: Student Identity Profile & Multi-Linked Email Settings Modal

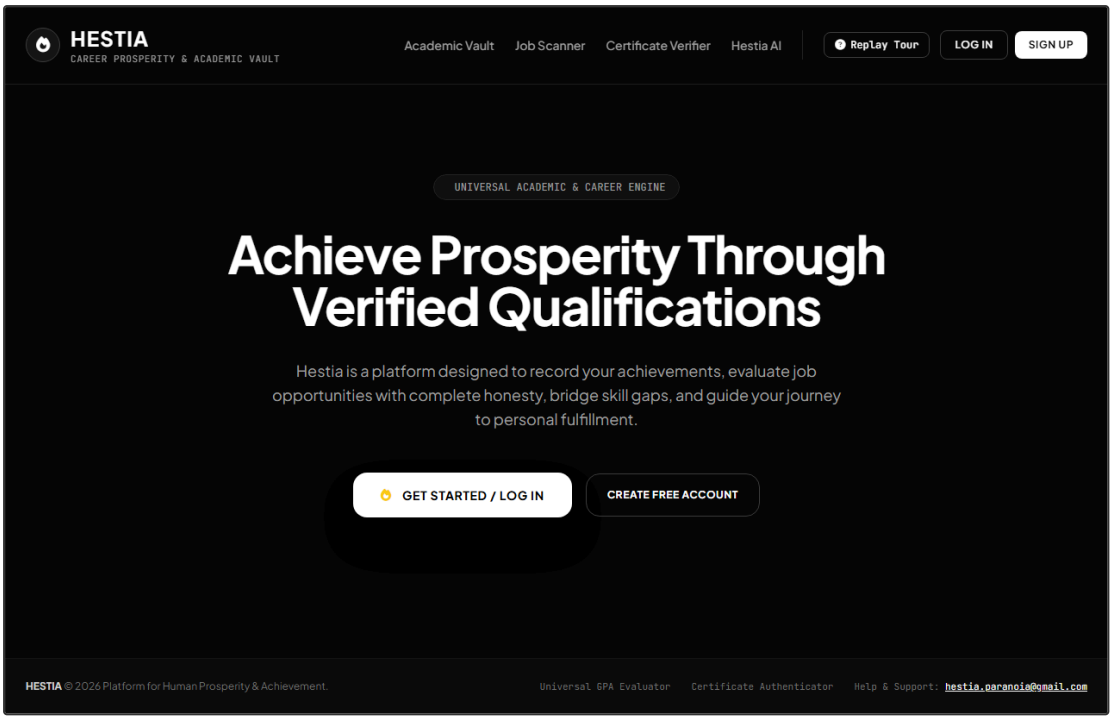


Fig 6.3: Universal Academic Landing Interface & Feature Navigation

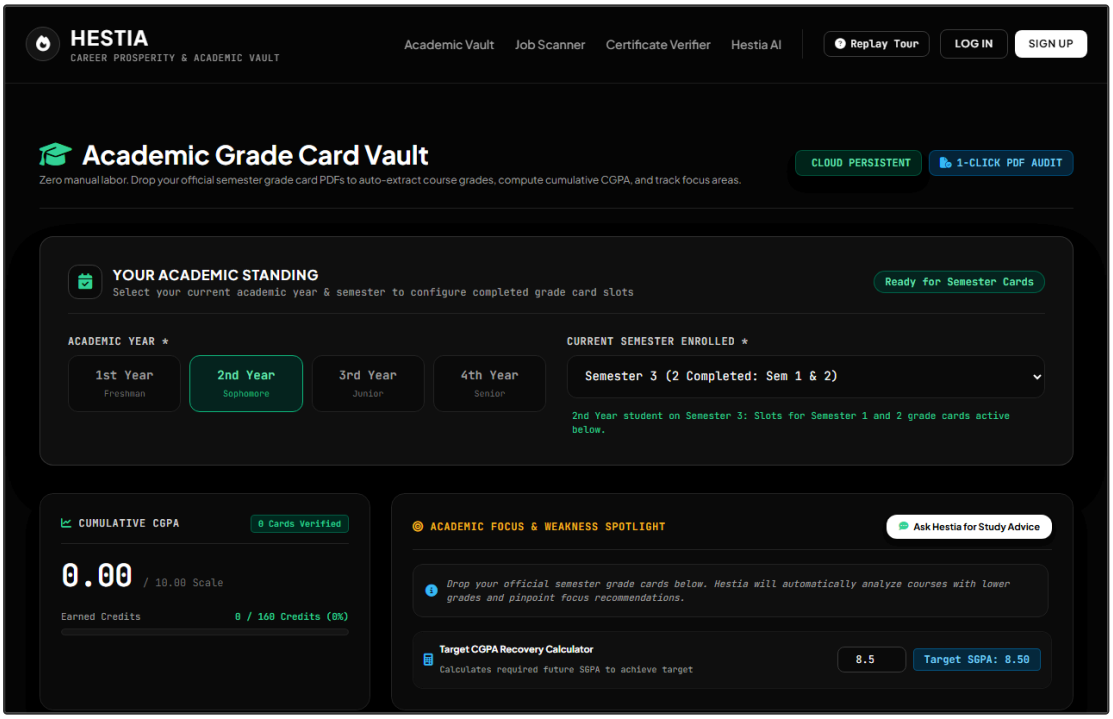


Fig 6.4: Academic Grade Card Vault & Target CGPA Calculator

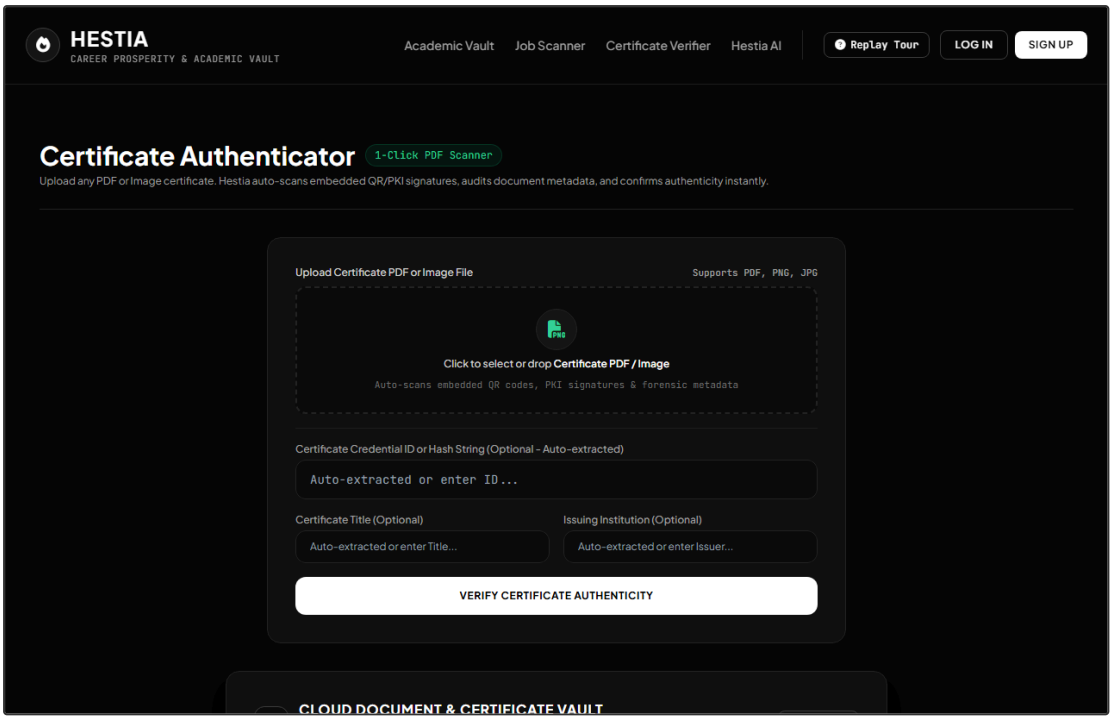


Fig 6.5: 1-Click PDF Certificate Authenticator & QR Validation Interface

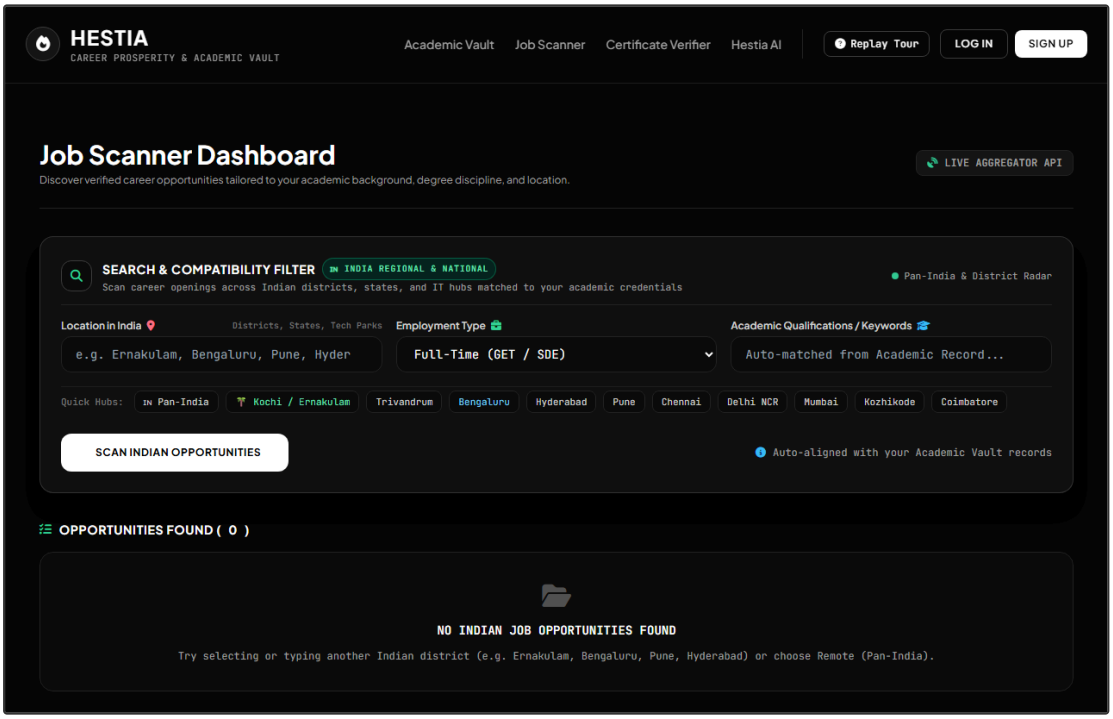


Fig 6.6: Pan-India District Job Radar & LPA Salary Matching

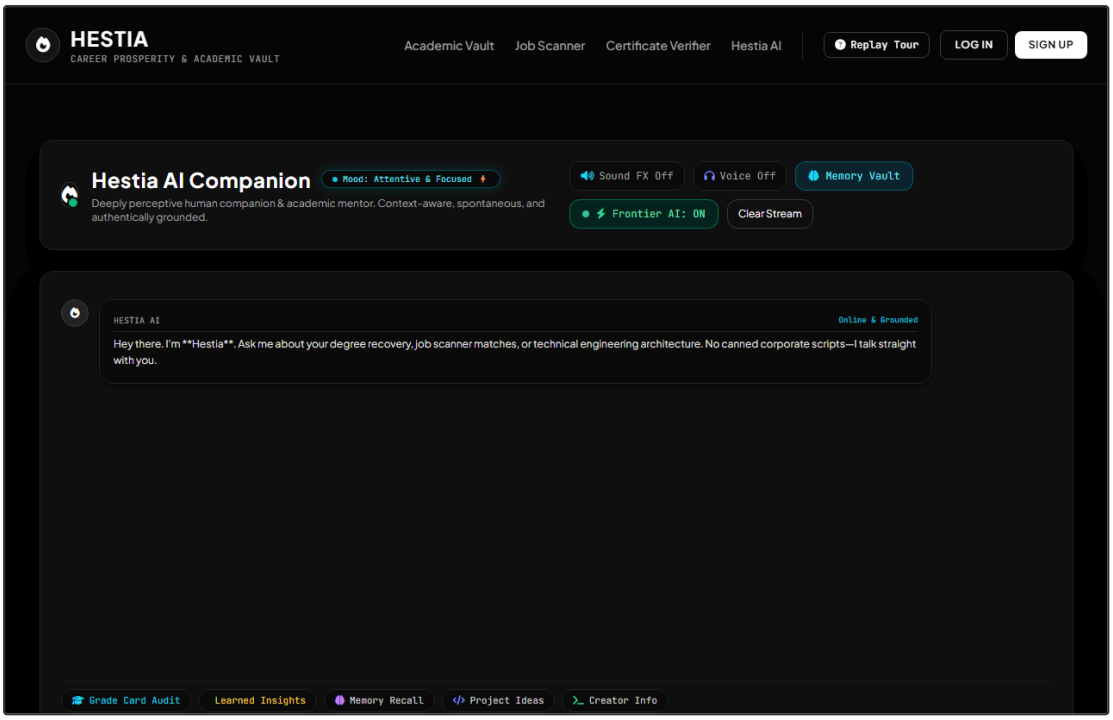


Fig 6.7: Hestia Frontier AI Career Companion & Neural Chatbot

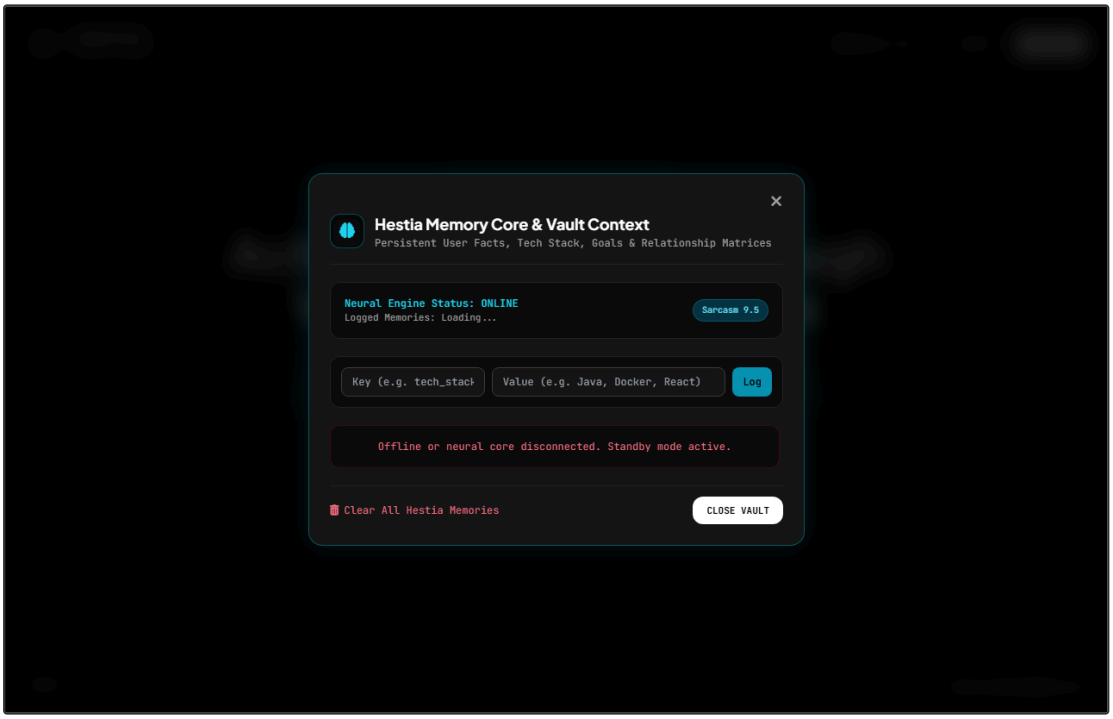


Fig 6.8: Hestia Memory Core Modal & Extracted Fact Vault